

PROBABILITY THEORY

MTM2592-Gr:2

THIRD WEEK

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DISCRETIZATION OF SAMPLE SPACE:

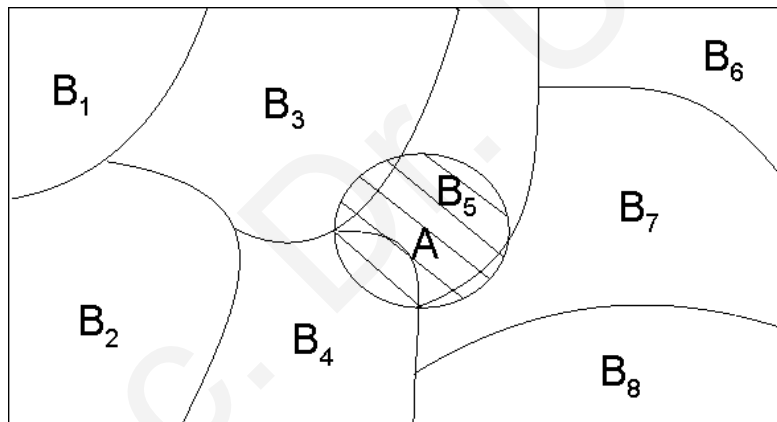
Let $B_i \subset S$,

$$a) \quad P(B_i \cap B_j) = \emptyset \quad i \neq j$$

$$b) \quad \bigcup_{i=1}^n B_i = S$$

$$c) \quad P(B_i) > 0 \quad \text{for all } i$$

Then $B_1, B_2, \dots, B_n$ events show a discretization of the sample space S .
The figure shows discretization of S for $n = 8$.



Theorem: According to the definition given above,

$$P(B_1) + P(B_2) + \dots + P(B_n) = 1$$

Theorem: If B_i ($i=1,2,\dots,n$) is a discretization of a sample space S , then

$$P(A) = \sum_{i=1}^n P(B_i \cap A) = \sum_{i=1}^n P(B_i) \cdot P(A|B_i)$$

for any events A in S .

Proof:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)$$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n)$$

$$P(A \cap B_i) = P(B_i)P(A|B_i)$$

$$P(A) = P(B_1)P(A|B_1) + \dots + P(B_n)P(A|B_n)$$

Since $A \cap B_i$
are disjoint events

Example: A type of machine is produced in the factories A, B and C. The number of machines produced in A is 2 times of machines produced in B and also in C. It is known that 2% of machines produced in A as well as in B, and 4% of machines produced in C are defective. If a machine is randomly selected from the store in which all produced machines put, what is the probability that the machine is defective?

Solution:

INDEPENDENT EVENTS:

Definition: The events A and B are independent if and only if

$$P(B|A) = P(B) \text{ or } P(A|B) = P(A)$$

Otherwise, A and B are dependent.

Theorem: The events A and B independent if and only if

$$P(A \cap B) = P(A).P(B)$$

Theorem: If A and B are independent events that $(P(A) \neq 0 \text{ and } P(B) \neq 0)$ then A and B have at least one common sample point. So, $A \cap B \neq \emptyset$

Theorem: If A is independent of B, then B is independent of A.

$$P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B)$$

Proof:
$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(A|B).P(B) = P(B|A).P(A)$$

$$P(A|B) = P(A), \quad P(A) \neq 0$$

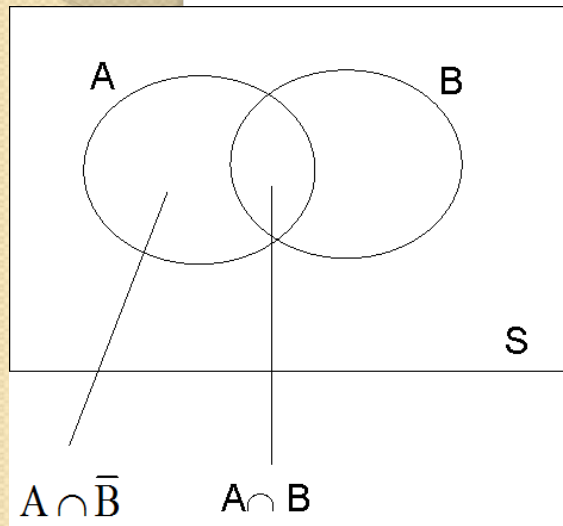
$$P(B) = P(B|A)$$

Theorem: If A and B are independent, then A and $\bar{B}$, $\bar{A}$ and B; $\bar{A}$ and $\bar{B}$ are also independent.

Proof: If A and B are independent, then A and $\bar{B}$ are also independent

$$A, B \subset S$$

$$P(A|B) = P(A) \Rightarrow P(A|\bar{B}) = P(A)$$



$$A = (A \cap B) \cup (A \cap \bar{B})$$

$$(A \cap B) \cap (A \cap \bar{B}) = \emptyset$$

$$P((A \cap B) \cup (A \cap \bar{B})) = P(A \cap B) + P(A \cap \bar{B})$$

$$P(A) = P(A)P(B) + P(A \cap \bar{B})$$

$$P(A)[1 - P(B)] = P(A \cap \bar{B})$$

$$P(A)P(\bar{B}) = P(A \cap \bar{B})$$

And hence A and $\bar{B}$ are independent.

TOTAL INDEPENDENCY

Definition: Events $A_1, A_2, \dots, A_n$ are total independent if and only if any $2, 3, \dots, (n-1)$ of these events are independent and

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

For example, if $n=3$,

$$P(A_1 \cap A_2 \cap A_3) = P(A_1).P(A_2).P(A_3)$$

$$P(A_1 \cap A_2) = P(A_1).P(A_2)$$

$$P(A_1 \cap A_3) = P(A_1).P(A_3)$$

$$P(A_2 \cap A_3) = P(A_2).P(A_3)$$

} for total independency

Note: $P(A_1 \cap \bar{A}_2 \cap A_3) = P(A_1).P(\bar{A}_2).P(A_3)$

Example: It is estimated that in Turkey, 22% of bachelors are woman and 15% of bachelors have economy degree. On the other hand, 7% of bachelors have economy degree and woman. Decide whether the events «be bachelor woman» and «be bachelor having economy degree» are independent or not.

Solution:

Example:

Figure 8 shows a Venn diagram with probabilities assigned to its various regions. Verify that A and B are independent, A and C are independent, and B and C are independent, but A , B , and C are not independent.

Solution

As can be seen from the diagram, $P(A) = P(B) = P(C) = \frac{1}{2}$, $P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4}$, and $P(A \cap B \cap C) = \frac{1}{4}$. Thus,

$$P(A) \cdot P(B) = \frac{1}{4} = P(A \cap B)$$

$$P(A) \cdot P(C) = \frac{1}{4} = P(A \cap C)$$

$$P(B) \cdot P(C) = \frac{1}{4} = P(B \cap C)$$

but

$$P(A) \cdot P(B) \cdot P(C) = \frac{1}{8} \neq P(A \cap B \cap C)$$

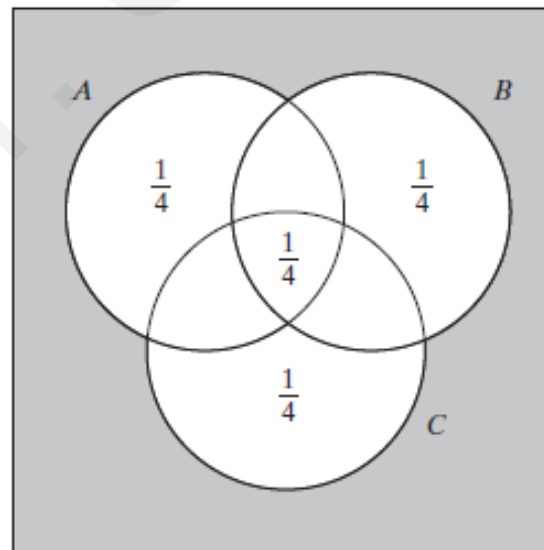


Figure 8. Venn diagram :

Example: Let a coin be tossed two times.

$A = \{\text{The coin shows tail at the first tossing}\}$

$B = \{\text{The coin shows tail at the second tossing}\}$

$C = \{\text{The coin shows tail only one times}\}$

Examine the total independency.

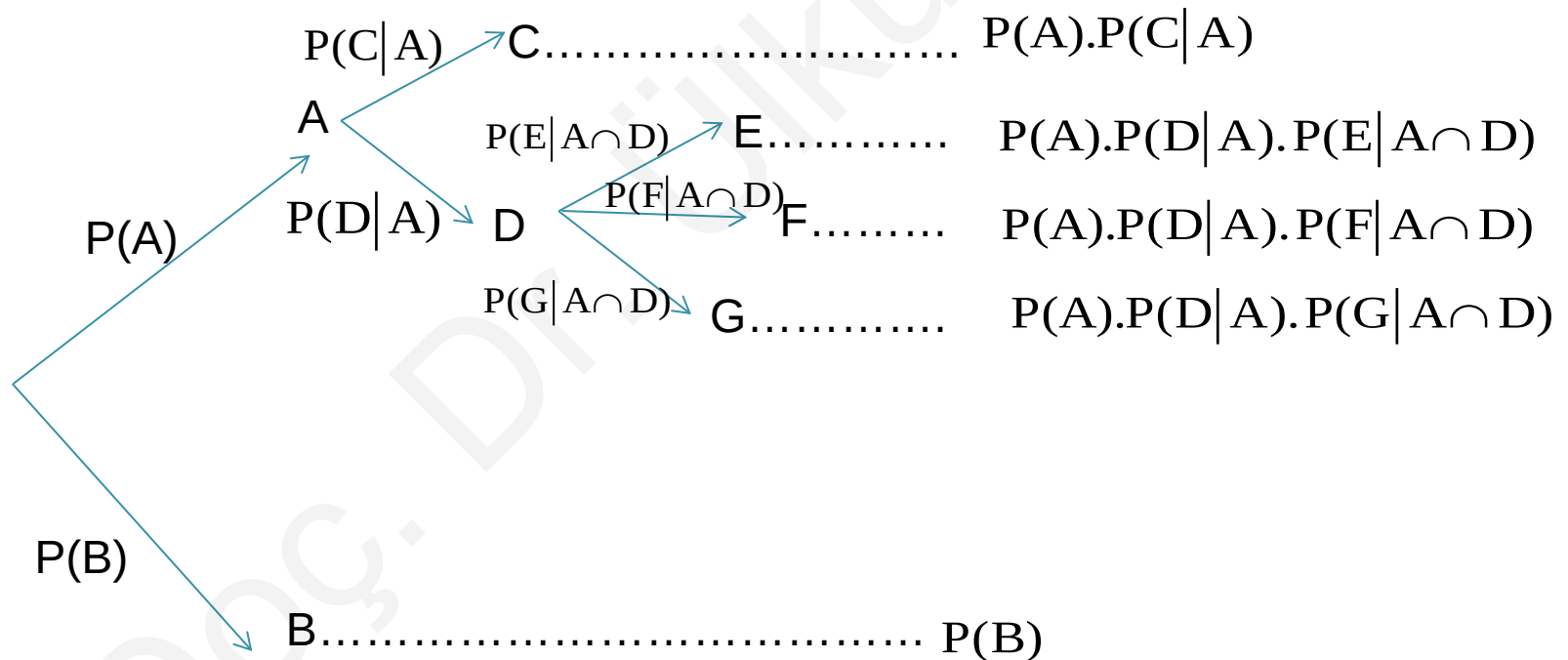
Solution:

Example: The probability of hitting a target is 95% for the person A and 82% for the person B. What is the probability that the target will be hit if both person try to hit the same target?

Solution:

DETERMINATION OF CONDITIONAL PROBABILITY BY TREE DIAGRAMS:

All terminated branches of a tree diagram are disjoint events and form the sample space of that experiment. While the probabilities in the first branches of the tree give the initial probabilities, the latter give conditional probabilities, the multiplication of the probabilities in the branches will give the intersection probabilities. In case the trials are independent, all the probabilities in the branches turn into unconditional probabilities, while the multiplication of the probabilities in the branches again gives the intersection probability.



Example: There are 3 white balls and 5 red balls in a box. 2 balls are successively selected from the box, without replacing them back. Find the probability that two balls are different coloured by using tree diagram.

Solution:

Example: There are 4 black and 6 red balls in the first box and 3 black and 2 red balls in the second box. If we randomly select a box, then randomly select a ball from the chosen box, what is the probability that

- a) The chosen ball is black,
- b) The chosen box is the first one given that a black ball is selected?

Solution:

BAYES' THEOREM:

Let $B_1, B_2, \dots, B_n$ be discretization of a sample space S and A be an event in S with $P(A) \neq 0$

Then,

$$P(B_r | A) = \frac{P(B_r) \cdot P(A | B_r)}{\sum_{i=1}^n P(B_i) P(A | B_i)}$$

Proof: Note:

$$P(A) = \sum_{i=1}^n P(B_i \cap A) = \sum_{i=1}^n P(B_i) \cdot P(A | B_i)$$

$$P(B_r | A) = \frac{P(A \cap B_r)}{P(A)} = \frac{P(B_r) \cdot P(A | B_r)}{\sum_{i=1}^n P(B_i) P(A | B_i)}$$

Note: Bayes' Theorem is based on obtaining an unknown $P(B | A)$ conditional probability using the known $P(A | B)$ conditional probability.

Example: In a factory, three machines B_1 , B_2 , B_3 , make 30%, 45% and 25%, respectively of the products. It is known that 2%, 3% and 2% of the products made by each machine, respectively are defective.

- a) What is the probability that a randomly selected product is defective?
- b) If a randomly selected product is defective, what is the probability that it was made by machine B_3 ?

Solution:

Example: Members of a consulting firm rent cars from three companies, 60% from Company 1, 30% from Company 2, and 10% from Company 3. If 9% of vehicles from Company 1, 20% of vehicles from Company 2 and 6% of vehicles from Company 3 require maintenance,

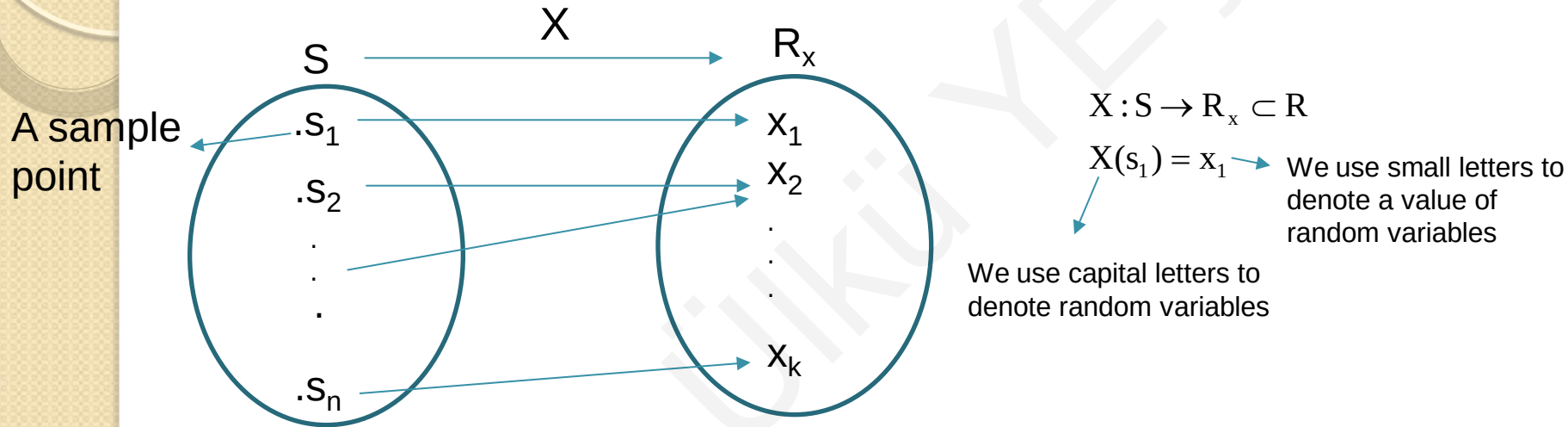
a) What is the probability that a car rented to the company will require maintenance?

b) If the car rented to the company requires maintenance, what is the probability that this vehicle came from Company 2?

Solution:

RANDOM VARIABLE

Definition: The random variable is a function from the sample space to the set of real numbers.



Ex: Rolling a die,

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$R_x = \{1, 2, 3, 4, 5, 6\}$$

Ex: If a coin tossed until it shows head and the random variable is the number of trials, then,

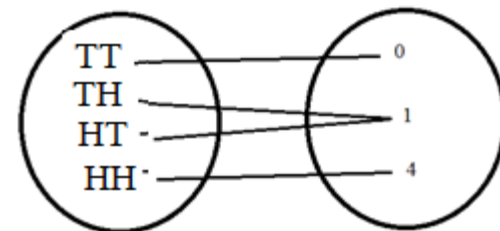
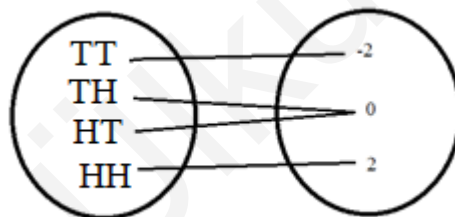
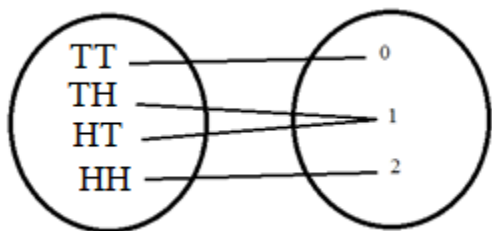
$$S = \{H, TH, TTH, TTTH, \dots\}$$

$$R_x = \{1, 2, 3, 4, \dots\}$$

Ex: A coin is tossed two times. Let random variable X shows the number of heads, random variable Y shows the difference of the number of heads from the number of tails, and random variable Z shows the square of the number of heads. Create definition and image sets of these random variables.

Definition Set $S = \{TT, TY, YT, YY\}$

Image Sets: $X(S) = \{0, 1, 2\}$, $Y(S) = \{-2, 0, 2\}$, $Z(S) = \{0, 1, 4\}$



Ex: Let two coin are tossed. Random variable Y shows the number of heads. There are 4 possible results

$$P\{Y = 0\} = P\{(T, T)\} = \frac{1}{4}$$

$$P\{H = 0\} + P\{H = 1\} + P\{H = 2\} = 1 \text{ and } P(Y = y) \geq 0$$

$$P\{Y = 1\} = P\{(H, T), (T, H)\} = \frac{2}{4}$$

$$P\{Y = 2\} = P\{(H, H)\} = \frac{1}{4}$$

Ex: Rolling of a pair of dies.

- a) X : the sum of numbers
- b) X : the absolute value of the difference of numbers

$$S = \{(x_1, x_2) : x_i = 1, 2, 3, 4, 5, 6; i = 1, 2\}$$

$$a) R_x = \{X : x = x_1 + x_2\} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$b) R_x = \{X : x = |x_1 - x_2|\} = \{0, 1, 2, 3, 4, 5\}$$

Ex: Let a string with 100 cm be cut randomly and X represents the length of longer piece.

$$R_x = \{X : 50 \leq x \leq 100\}$$

If R_x that contains value of the random variable X is finite or countably infinite, then X is said to be discrete. If R_x is an interval or a set of intervals, then X is said to be continuous.

Ex: Two socks were randomly selected from a drawer containing 5 brown and 3 green socks, successively. List the elements of sample space, the corresponding probabilities and the corresponding values w of the random variable W , where W the number of brown socks selected.

Solution:

Sample Space	Probability	w
BB	$\frac{5}{8} \cdot \frac{4}{7} = \frac{5}{14}$	2
BG	$\frac{5}{8} \cdot \frac{3}{7} = \frac{15}{56}$	1
GB	$\frac{3}{8} \cdot \frac{5}{7} = \frac{15}{56}$	1
GG	$\frac{3}{8} \cdot \frac{2}{7} = \frac{3}{28}$	0

Also, we can write $P(W = 2) = \frac{5}{14}$, for example, for the probability of the event that the random variable W will take on the value 2.

Ex: A balanced coin is tossed four times. List the elements of the sample space that are presumed to be equally likely, as this is what we mean by a coin being balanced, and the corresponding values x of the random variable X , the total number of heads.

Solution

If H and T stand for heads and tails, the results are as shown in the following table:

<i>Element of sample space</i>	<i>Probability</i>	<i>x</i>	<i>Element of sample space</i>	<i>Probability</i>	<i>x</i>
HHHH	$\frac{1}{16}$	4	HTTH	$\frac{1}{16}$	2
HHHT	$\frac{1}{16}$	3	THHT	$\frac{1}{16}$	2
HHTH	$\frac{1}{16}$	3	THTH	$\frac{1}{16}$	2
HTHH	$\frac{1}{16}$	3	TTHH	$\frac{1}{16}$	2
THHH	$\frac{1}{16}$	3	HTTT	$\frac{1}{16}$	1
HHTT	$\frac{1}{16}$	2	THTT	$\frac{1}{16}$	1
HTHT	$\frac{1}{16}$	2	TTHT	$\frac{1}{16}$	1
			TTTH	$\frac{1}{16}$	1
			TTTT	$\frac{1}{16}$	0

Thus, we can write $P(X = 3) = \frac{4}{16}$, for example, for the probability of the event that the random variable X will take on the value 3.

Example:

Find a formula for the probability distribution of the total number of heads obtained in four tosses of a balanced coin.

Solution

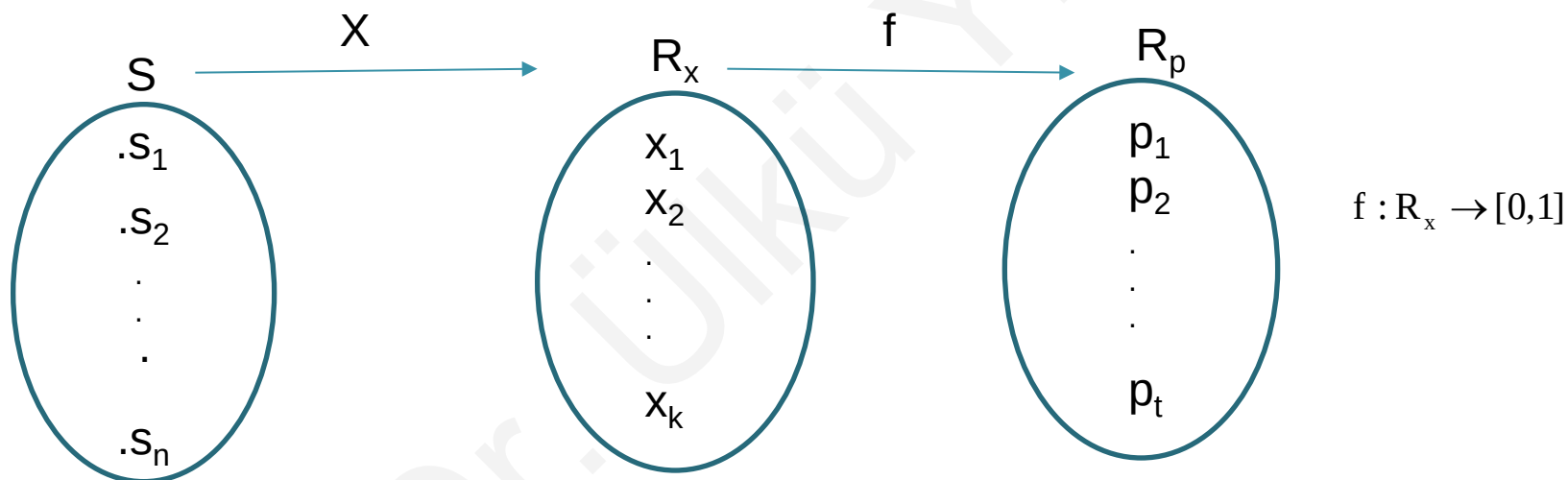
Based on the probabilities in the table, we find that $P(X = 0) = \frac{1}{16}$, $P(X = 1) = \frac{4}{16}$, $P(X = 2) = \frac{6}{16}$, $P(X = 3) = \frac{4}{16}$, and $P(X = 4) = \frac{1}{16}$. Observing that the numerators of these five fractions, 1, 4, 6, 4, and 1, are the binomial coefficients $\binom{4}{0}$, $\binom{4}{1}$, $\binom{4}{2}$, $\binom{4}{3}$, and $\binom{4}{4}$, we find that the formula for the probability distribution can be written as

$$f(x) = \frac{\binom{4}{x}}{16} \quad \text{for } x = 0, 1, 2, 3, 4$$

PROBABILITY FUNCTION

1)PROBABILITY FUNCTION OF A DISCRETE RANDOM VARIABLE:

Definition: Let $X(S) = R_x = \{x_1, x_2, \dots, x_k\}$ be given



The function $f(x_i) = P(X = x_i)$, $i = 1, 2, \dots, k$ which represents the probability that the random variable X is the value x_i is called probability function or probability ditribution.

Based on the postulates of probability, we obtain the following theorem.

THEOREM 1. A function can serve as the probability distribution of a discrete random variable X if and only if its values, $f(x)$, satisfy the conditions

1. $f(x) \geq 0$ for each value within its domain;
2. $\sum_x f(x) = 1$, where the summation extends over all the values within its domain.

Ex: A probability function of X random variable is given in table,

x_i	1	2	3	4
$f(x_i)$	1/4	1/5	1/20	1/2

then obtain $Y=3X+1$ random variable's probability function.

Solution:

y_i	4	7	10	13
$f(y_i)$	1/4	1/5	1/20	1/2

Ex: The probability function of random variable X is as follows.

$$f(x) = \begin{cases} -\frac{1}{3}(x-2) & x = 0, 1, 2 \\ 0 & \text{Otherwise} \end{cases}$$

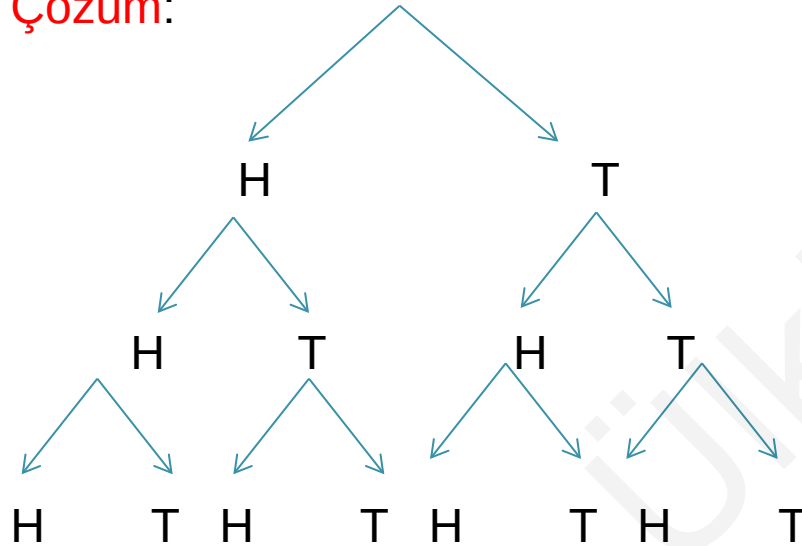
What is the probability function of the random variable Y, defined as $Y = 2X$?

Solution: $Y=2X$ or $X=Y/2$

$$f(y) = P(Y = y) = \begin{cases} -\frac{1}{3}\left(\frac{y}{2} - 2\right) & y = 0, 2, 4 \\ 0 & \text{Otherwise} \end{cases}$$

Ex: 3 coins are tossed together. If the number of heads are represented by the random variable X , determine the probability function.

Çözüm:



$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$R_x = \{0, 1, 2, 3\}$$

X	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

$$f(x) = \begin{cases} 1/8, & x = 0 \\ 3/8, & x = 1 \\ 3/8, & x = 2 \\ 1/8, & x = 3 \end{cases}$$

Ex: Determine that $f(x) = \frac{2x}{k(k+1)}$, $x = 1, 2, \dots, k$ is a probability function?

Solution:

Ex: Let $f(x) = \begin{cases} 2kx, & x = 1, 2, 3 \\ k(3+x), & x = 4, 5 \\ 0, & \text{otherwise} \end{cases}$

- a) Find the value of k if f is a probability function,
- b) Compute $P(X=8)$, $P(X=3)$ and $P(x \geq 2 | x \leq 4)$

Ex: If X is defined as the number of boys in a family with 3 children, find the probability function of X . Calculate the probability that the family has at least one boy.

2) PROBABILITY FUNCTION OF A CONTINUOUS RANDOM VARIABLE

Definition: The function f is called the probability (density) function of the continuous random variable X , if

1) $f(x) \geq 0$ for all x values

2) $\int_{-\infty}^{\infty} f(x) dx = 1$

are satisfied.

The probability $P(a \leq X \leq b) = \int_a^b f(x) dx$

For a constant a

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x) dx = 0 \quad \Rightarrow \quad \text{It means that the probability of a single point is zero for continuous variables}$$

Thus,

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$

Ex: If $f(x) = \begin{cases} kx^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$

is a probability function, find the value of k.

Solution:

CUMULATIVE DISTRIBUTION (DISTRIBUTION FUNCTION):

Definition: The cumulative distribution $F(x)$ of a **discrete random** variable X with probability function $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{u \leq x} f(u) \quad \text{for } -\infty < x < \infty$$

If X is a **continuous random** variable, then

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \quad \text{for } -\infty < x < \infty$$

Ex:

x	0	1	2
f(x)	0.15	0.50	0.35

Determine the cumulative function F and plot it.

Solution:

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.15, & 0 \leq x < 1 \\ 0.65, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

Some Properties of Cumulative Distribution of a Discrete Variable

a) $F(x)$; is a non-decreasing step function in the interval $[0,1]$

b) $P(X > a) = 1 - P(X \leq a) = 1 - F(a)$

c) $P(a < X \leq b) = F(b) - F(a)$

Some Properties of Cumulative Distribution of a Continuous Variable

a) $\lim_{x \rightarrow -\infty} F(x) = F(-\infty) = 0$

b) $\lim_{x \rightarrow \infty} F(x) = F(\infty) = 1$

c) $F(x)$; is a non-decreasing step function in the interval $[0,1]$.

d) $\frac{d}{dx} F(x) = f(x)$ $(F'(x) = f(x))$

e) $P(X > a) = 1 - F(a)$ f) $P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x) dx$

Ex:

Find the distribution function of the total number of heads obtained in four tosses of a balanced coin.

Solution

Given $f(0) = \frac{1}{16}, f(1) = \frac{4}{16}, f(2) = \frac{6}{16}, f(3) = \frac{4}{16}$, and $f(4) = \frac{1}{16}$ it follows that

$$F(0) = f(0) = \frac{1}{16}$$

$$F(1) = f(0) + f(1) = \frac{5}{16}$$

$$F(2) = f(0) + f(1) + f(2) = \frac{11}{16}$$

$$F(3) = f(0) + f(1) + f(2) + f(3) = \frac{15}{16}$$

$$F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1$$

Hence, the distribution function is given by

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{16} & \text{for } 0 \leq x < 1 \\ \frac{5}{16} & \text{for } 1 \leq x < 2 \\ \frac{11}{16} & \text{for } 2 \leq x < 3 \\ \frac{15}{16} & \text{for } 3 \leq x < 4 \\ 1 & \text{for } x \geq 4 \end{cases}$$

Observe that this distribution function is defined not only for the values taken on by the given random variable, but for all real numbers. For instance, we can write $F(1.7) = \frac{5}{16}$ and $F(100) = 1$, although the probabilities of getting “at most 1.7 heads” or “at most 100 heads” in four tosses of a balanced coin may not be of any real significance.

Example: Let X be a random variable that indicates the charging time of a battery. If the probability density function;

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2} & x > 0 \\ 0 & \text{Other Cases} \end{cases}$$

- What is the probability that this battery will charge in less than 1 hour or more than 3 hours?
- What is the probability that this time will be longer than 2 hours given that the battery is known to be charged in more than 1 hour?
- What is F(x)?

Solution: A: Charging time less than 1 hour
B: Charging time is more than 3 hours

a)

$$P(A \cup B) = P(A) + P(B)$$

$$= \int_0^1 \frac{1}{2}e^{-x/2} dx + \int_3^{\infty} \frac{1}{2}e^{-x/2} dx$$

$$= 1 - e^{-1/2} + e^{-3/2}$$

$$= 1 - 0,61 + 0,22$$

$$= 0,61$$

b)

$$b) P(X > 2 \mid X > 1) = \frac{P(X > 2)}{P(X > 1)}$$

$$\frac{P(X > 2)}{P(X > 1)} = \frac{\int_2^{\infty} \frac{1}{2}e^{-x/2} dx}{\int_1^{\infty} \frac{1}{2}e^{-x/2} dx} = \frac{e^{-1}}{e^{-1/2}} = e^{-1/2} = 0,61$$

c)

$$\begin{aligned} \text{c) } F(x) &= P(X \leq x) = \int_0^x \frac{1}{2} \cdot e^{-\frac{x}{2}} dx \\ &= \frac{1}{2}(-2)e^{-x/2} \Big|_0^x \\ &= 1 - e^{-x/2} \end{aligned}$$

$$F(x) = \begin{cases} 1 - e^{-x/2} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Example: Cumulative function of X is given by

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x(2-x) & 0 < x < 1 \\ 1 & 1 \leq x \end{cases}$$

If $Y=2X+1$, then find the cumulative function of $G(y)$ and probability density function $g(y)$.

Solution:

$$\begin{aligned} G(y) &= P(Y \leq y) = P(2X+1 \leq y) \\ &= P\left(x \leq \frac{y-1}{2}\right) \\ &= F\left(\frac{y-1}{2}\right) \end{aligned}$$

$$G(y) = \begin{cases} 0 & y \leq 1 \\ \frac{y-1}{2} \left(2 - \frac{y-1}{2} \right) = \frac{-y^2 + 6y - 5}{4} & 1 < y < 3 \\ 1 & 3 \leq y \end{cases}$$

$$G'(y) = g(y)$$

$$g(y) = \frac{dG(y)}{dy} = \frac{d\left[(-y^2 + 6y - 5)/4\right]}{dy} = \frac{1}{4}(-2y + 6)$$

$$g(y) = \frac{1}{2}(-y + 3)$$

$$g(y) = \begin{cases} -\frac{y}{2} + \frac{3}{2}, & 1 < y < 3 \\ 0, & \text{elsewhere} \end{cases}$$

Ex: If $F(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$, find $f(x)$.

Solution:

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases} \quad \text{For } 0 < x < 1 \quad \frac{dF}{dx} = f(x) \Rightarrow \frac{d(x)}{dx} = 1 = f(x)$$

Example: Probability function of discrete random variable of X is given below;

$X=x:$	-3	0	2	3
$P(X=x):$	0,2	0,1	0,4	c

- Find the value of c?
- At which value, X's probability maximum?
- $P(X>0)=?$
- $P(X=-2)=?$
- Find the cumulative function of X?
- Show F(x) as a Table.
- Draw the Graphs of f(x) and F(x)

Solution:

a) $0,2+0,1+0,4+c=1 \implies c=0,3$

b) 2

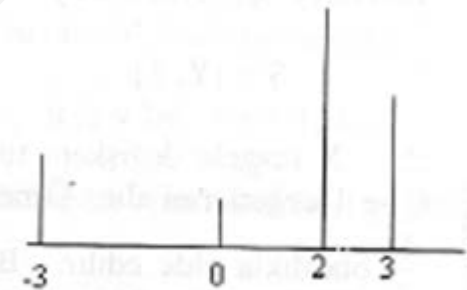
c) $P(X>0) = P(X=2)+P(X=3)=0,4+0,3=0,7.$

d) $P(X = -2) = 0.$

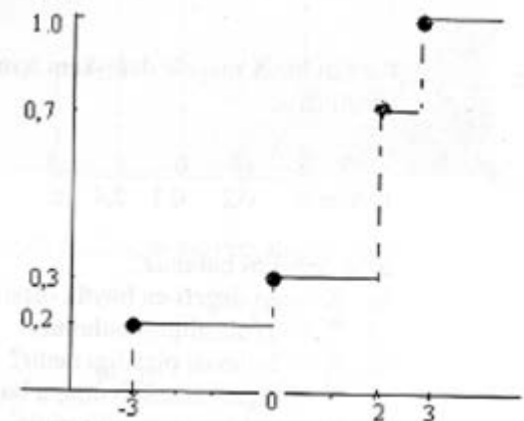
e)
$$F(x) = \begin{cases} 0 & x < -3 \\ 0,2 & -3 \leq x < 0 \\ 0,3 & 0 \leq x < 2 \\ 0,7 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

f)

$X=x:$	-3	0	2	3
$F(x)=P(X \leq x):$	0,2	0,3	0,7	1.0



Graphs of f(x)



Graphs of cumulative function of F(x)

Example: A coin is tossed until a head occurs. Let X be a random variable that shows necessary number of tossing. Find the probability function of random variable X.

Solution:

		Probability
First toss	H	1/2
Second toss	TH	1/2.1/2
Third toss	TTH	1/2.1/2.1/2
.	.	.
N-th toss	TTT...TH	$(1/2)^{N-1}.1/2$
.	.	.

X = x	1	2	...	x	...
$f(x) = P(X=x)$	$\frac{1}{2}$	$\left(\frac{1}{2}\right)^2$	...	$\left(\frac{1}{2}\right)^x$	...

$$f(x) = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, \dots$$

Note: $a + ar + ar^2 + \dots = \frac{a}{1-r}$

$$(-1 \leq r \leq 1)$$

$$\sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^x = \frac{1}{2} + \frac{1}{2^2} + \dots$$

$$a = \frac{1}{2}, r = \frac{1}{2} \Rightarrow \sum_{x=1}^{\infty} f(x) = \frac{1/2}{1-1/2} = 1$$